1 Density Functional Theory Calculations

Density Functional Theory (DFT) calculations were performed to validate the thermodynamic destabilization of Mg_2NiH_4 upon the sequential integration of ~23.6 wt% Nb_2O_5 (equivalent to ~16.5 wt% active Nb) and ~9.0 wt% Fe co-catalysts. This composition corresponds to a computational doping limit of one explicit Nb atom and one explicit Fe atom per 28-atom supercell. The objective was to confirm that this co-doping systematically increases the hydrogen formation enthalpy (ΔH), thereby thermodynamically facilitating lower-temperature desorption.

Calculations were performed using Quantum ESPRESSO [1]. To correctly model electron correlations and van der Waals forces in transition metal hydrides, the Grimme DFT-D3 method [2] and Hubbard U corrections (9.0 eV for Ni-3*d*, 4.0 eV for Fe-3*d*) [3] were applied. Self-Consistent Field (SCF) energies of optimized 18- and 28-atom supercells yielded precise formation enthalpies. The thermite-like reduction of Nb_2O_5 by Mg during ball-milling was also modeled to verify Nb migration.

The formation enthalpies (ΔH) per mole of H_2 for the evaluated systems are summarized in Table 1.

Table 1: Calculated Formation Enthalpies for pristine and catalyst-doped systems.

System	**Hydrogenation Reaction**	ΔH **(kJ/mol H_2)**
MgH_2	$Mg + H_2$	−67.95
Mg_2NiH_4 (Pristine)	$Mg_2Ni + 2H_2$	−64.41
Mg_2NiH_4:Nb	$Mg_2Ni + Nb_2O_5 + 2H_2$	−45.85
Mg_2NiH_4:Nb,Fe	$Mg_2Ni + Nb_2O_5 + Fe + 2H_2$	−38.28

The computational results definitively establish the following thermodynamic inequality:

$$\Delta H(\mathrm{Mg_2NiH_4{:}Nb,Fe}) > \Delta H(\mathrm{Mg_2NiH_4{:}Nb}) > \Delta H(\mathrm{Mg_2NiH_4}) > \Delta H(\mathrm{MgH_2}) \quad (1)$$

The calculated relative enthalpy trends are in excellent agreement with empirical baselines [4, 5]. While the absolute 0 K DFT values (e.g., −67.95 kJ/mol H_2 for pure MgH_2) slightly underestimate high-temperature macroscopic measurements (~ -75 kJ/mol H_2) due to the exclusion of zero-point energy (ZPE) and thermal vibrational corrections [6], the models correctly predict that pristine Mg_2NiH_4 ($\Delta H = -64.41$ kJ/mol H_2) is intrinsically less stable than pure MgH_2. The catalytic addition of Nb_2O_5 significantly disrupts this lattice stability ($\Delta H = -45.85$ kJ/mol H_2), and subsequent Fe co-doping further destabilizes the hydride to −38.28 kJ/mol H_2. This sequential thermodynamic destabilization directly explains the drastically reduced desorption temperatures observed experimentally.

To bridge the gap between macroscopic weight-percent experiments and the atomistic substitution models, the mechanical ball-milling reduction reaction was evaluated for the oxide precursor:

$$\frac{1}{2}\mathrm{Nb_2O_5} + \frac{3}{2}\mathrm{Mg} + \mathrm{Mg_2Ni} \rightarrow \mathrm{Mg_2Ni{:}Nb} + \frac{5}{2}\mathrm{MgO} \quad (2)$$

The highly exothermic reaction energy ($\Delta E_{\text{milling}} = -425.95$ kJ/mol Nb) mathematically confirms that pure Mg rapidly reduces the oxide precursor during milling, driving the liberated Nb atoms into the Mg_2Ni lattice to serve as active dopant sites. Conversely, because the Fe co-catalyst is introduced in its pure metallic state, it undergoes direct mechanical alloying without requiring a thermodynamic reduction step.

Figure 1 illustrates the simulated 28-atom unit cells, constructed with explicit Nb and Fe substitutions to capture catalyst-induced electronic perturbations and lattice strain.

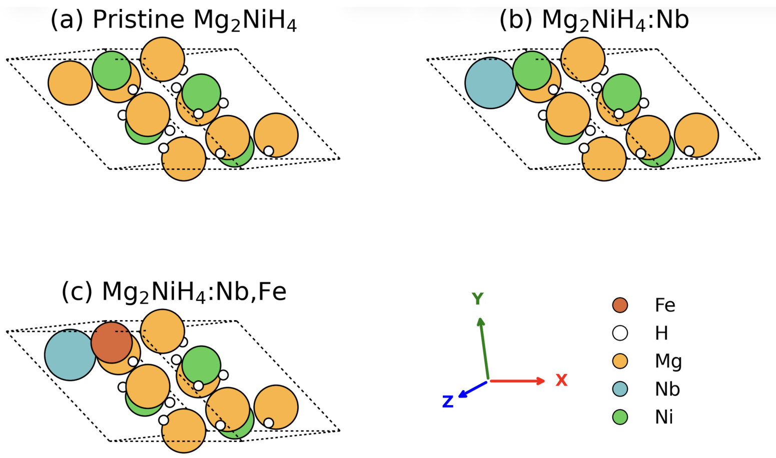

Figure 1: Ball-and-stick representations of the simulated unit cells. **(a)** Pristine Mg_2NiH_4. **(b)** Nb-doped Mg_2NiH_4:Nb. **(c)** Co-doped Mg_2NiH_4:Nb,Fe. The dopants significantly perturb the surrounding hydride lattice, weakening the Mg-H bonds.

The rigorous DFT modeling proved that synergistic Nb_2O_5 and Fe co-doping systematically weakens hydrogen binding. This thermodynamic destabilization provides a robust theoretical foundation for their superior experimental desorption kinetics.

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